A counterexample to uniform L^2 stability in the Brascamp–Lieb variance inequality

Abstract

We show that the sharp uniform L^1 stability estimate for the Brascamp–Lieb variance inequality cannot be strengthened to an L^2 estimate with a constant depending only on the dimension.

1 Statement

Let $\phi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be an essentially continuous convex function with $0 < \int e^{-\phi} < \infty$, and put

$$d\rho_\phi = Z_\phi^{-1} e^{-\phi}\, dx.$$

For smooth strictly convex ϕ, the Brascamp–Lieb deficit is

$$\delta_{\mathrm{BL},\phi}(f) = \int_{\mathbb{R}^d} \big\langle (D^2\phi)^{-1}\nabla f, \nabla f \big\rangle\, d\rho_\phi - \mathrm{Var}_{\rho_\phi}(f).$$

The equality cases are

$$\mathcal{O}_{\mathrm{BL},\phi} = \{a \cdot \nabla\phi + b : \ a \in \mathbb{R}^d,\ b \in \mathbb{R}\}.$$

Theorem 1.1. *Already in dimension one, there is no finite constant C such that*

$$\mathrm{dist}_{L^2(\rho_\phi)}(f, \mathcal{O}_{\mathrm{BL},\phi}) \le C\, \delta_{\mathrm{BL},\phi}(f)^{1/2}$$

for every smooth strictly convex $\phi : \mathbb{R} \to \mathbb{R}$ with $\int e^{-\phi} < \infty$ and every locally Lipschitz $f \in L^2(\rho_\phi)$. Consequently no dimension-only constant can exist in any dimension $d \ge 1$.

The point is spectral. In one dimension the BL Dirichlet form is

$$\mathcal{E}_\phi(f) = \int_{\mathbb{R}} \frac{|f'|^2}{\phi''}\, d\rho_\phi.$$

The function ϕ' is always an eigenfunction with eigenvalue 1. We construct potentials for which there is a second, even, almost-eigenfunction with Rayleigh quotient arbitrarily close to 1. Since this even mode is orthogonal to the constants and to the odd function ϕ', it is far from the optimizer space compared with its deficit.

2 The construction

Fix a non-negative even C^∞ bump κ supported in $[-1,1]$ with $\int_{\mathbb{R}} \kappa = 1$. For parameters $S \gg 1$, $\varepsilon > 0$, and $\eta > 0$, define a smooth even strictly convex potential by prescribing

$$\phi''_{S,\varepsilon,\eta}(x) = \eta + \frac{S}{\varepsilon}\kappa\left(\frac{x-1}{\varepsilon}\right) + \frac{S}{\varepsilon}\kappa\left(\frac{x+1}{\varepsilon}\right), \qquad \phi_{S,\varepsilon,\eta}(0) = \phi'_{S,\varepsilon,\eta}(0) = 0.$$

We choose $\varepsilon = \varepsilon(S)$ and $\eta = \eta(S)$ so small that

$$S\varepsilon \to 0, \qquad \eta S^{-2} \to 0, \qquad \eta \to 0.$$

For instance $\varepsilon = S^{-3}$ and $\eta = S^{-4}$ are enough. Write

$$\phi_S := \phi_{S,S^{-3},S^{-4}}, \qquad \rho_S := \rho_{\phi_S}, \qquad Z_S := \int_{\mathbb{R}} e^{-\phi_S}.$$

The potential is almost flat on $[-1,1]$, while its derivative jumps from nearly 0 to nearly S across each of the two small layers near ± 1. After the right layer, $\phi'_S = S + o(S)$ throughout the part of the right tail carrying essentially all of the tail mass. Hence

$$Z_S = 2 + \frac{2}{S} + o(S^{-1}), \qquad \rho_S((1+\varepsilon,\infty)) = \frac{1+o(1)}{Z_S S}. \tag{1}$$

The same estimate holds on the left by symmetry.

Now define an even Lipschitz test function g_S as follows. Let

$$I_S^+ = (1-\varepsilon, 1+\varepsilon), \qquad I_S^- = (-1-\varepsilon, -1+\varepsilon).$$

Set $g_S = 0$ on $[-1+\varepsilon, 1-\varepsilon]$ and $g_S = 1$ on $(-\infty, -1-\varepsilon] \cup [1+\varepsilon, \infty)$. Inside each transition layer choose the energy-minimising interpolation for the Dirichlet form $\mathcal{E}_{\phi_S}$ with these boundary values. Equivalently, on I_S^+,

$$g'_S(x) = \frac{\phi''_S(x) e^{\phi_S(x)}}{\int_{I_S^+} \phi''_S(t) e^{\phi_S(t)}\, dt},$$

and on I_S^- we take the symmetric interpolation.

Lemma 2.1. *With the above choice,*

$$\mathcal{E}_{\phi_S}(g_S) = \frac{2}{Z_S \int_{I_S^+} \phi''_S(t) e^{\phi_S(t)}\, dt} = \frac{2+o(1)}{Z_S S}.$$

Proof. For a single interval I and boundary jump Δ, the minimum of

$$Z_S^{-1} \int_I \frac{|h'|^2}{\phi''_S} e^{-\phi_S}\, dx$$

among all interpolants with jump Δ is

$$\frac{\Delta^2}{Z_S \int_I \phi_S'' e^{\phi_S}}.$$

This is the elementary one-dimensional resistance formula, obtained from the Euler–Lagrange equation $h'(\phi_S'')^{-1}e^{-\phi_S} = \text{constant}$. Here $\Delta = 1$ on both layers. Moreover $\phi_S = o(1)$ uniformly on I_S^+ because the layer width satisfies $S\varepsilon \to 0$, while

$$\int_{I_S^+} \phi_S'' = S + O(\eta\varepsilon) = S + o(S).$$

Thus $\int_{I_S^+} \phi_S'' e^{\phi_S} = S + o(S)$. □

Lemma 2.2. *The variance of g_S satisfies*

$$\mathrm{Var}_{\rho_S}(g_S) = \frac{2 + o(1)}{Z_S S}.$$

Proof. The two exterior tails have total ρ_S-mass

$$q_S = \rho_S((-\infty, -1-\varepsilon] \cup [1+\varepsilon, \infty)) = \frac{2 + o(1)}{Z_S S}$$

by (1). The transition layers have mass $o(S^{-1})$, since their length is $O(\varepsilon)$ and $e^{-\phi_S} = 1 + o(1)$ there. Hence g_S differs in $L^2(\rho_S)$ by $o(S^{-1/2})$ from the indicator of the two exterior tails. Therefore

$$\mathrm{Var}_{\rho_S}(g_S) = q_S(1 - q_S) + o(S^{-1}) = \frac{2 + o(1)}{Z_S S}.$$

□

3 Conclusion

Let

$$f_S = g_S - \int g_S \, d\rho_S.$$

Subtracting a constant does not change the BL energy or the variance, so

$$\delta_{\mathrm{BL},\phi_S}(f_S) = \mathcal{E}_{\phi_S}(g_S) - \mathrm{Var}_{\rho_S}(g_S).$$

By the Brascamp–Lieb inequality this quantity is non-negative, and Lemmas 2.1–2.2 give

$$\frac{\mathcal{E}_{\phi_S}(g_S)}{\mathrm{Var}_{\rho_S}(g_S)} \longrightarrow 1.$$

Equivalently,

$$\frac{\delta_{\mathrm{BL},\phi_S}(f_S)}{\mathrm{Var}_{\rho_S}(f_S)} = \frac{\mathcal{E}_{\phi_S}(g_S)}{\mathrm{Var}_{\rho_S}(g_S)} - 1 \longrightarrow 0.$$

Since ϕ_S is even, ϕ_S' is odd. The function f_S is even and has mean zero. Therefore f_S is orthogonal in $L^2(\rho_S)$ both to constants and to ϕ_S', i.e. to the whole one-dimensional optimizer space $\mathcal{O}_{\mathrm{BL},\phi_S} = \{a\phi_S' + b\}$. Thus

$$\mathrm{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2 = \|f_S\|^2_{L^2(\rho_S)} = \mathrm{Var}_{\rho_S}(g_S).$$

Consequently

$$\frac{\mathrm{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2}{\delta_{\mathrm{BL},\phi_S}(f_S)} = \frac{\mathrm{Var}_{\rho_S}(g_S)}{\mathcal{E}_{\phi_S}(g_S) - \mathrm{Var}_{\rho_S}(g_S)} \longrightarrow +\infty.$$

No universal L^2 stability constant can therefore exist. This proves Theorem 1.1 in dimension one.

For dimension $d > 1$, take the product potential

$$\Phi_S(x, y) = \phi_S(x) + \frac{|y|^2}{2}, \qquad (x, y) \in \mathbb{R} \times \mathbb{R}^{d-1},$$

and the test function $F_S(x, y) = f_S(x)$. The product structure leaves the variance and BL energy unchanged. Moreover F_S is orthogonal to constants, to the odd function $\phi_S'(x)$, and to all Gaussian coordinate functions $y_1, \ldots, y_{d-1}$. Hence its L^2 distance to the full optimizer space for Φ_S is still its L^2 norm, and the same divergent ratio proves the claim in every fixed dimension.

Remark 3.1. The earlier idea of using only an unbounded ordinary Poincare constant is not sufficient: the Brascamp–Lieb form is affinely invariant, while the Euclidean Poincare constant is not. The construction above instead creates an actual near-equality mode for the BL Rayleigh quotient outside the optimizer space.

References

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[2] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log-concave functions, and with an application to the diffusion equation*, J. Funct. Anal. 22 (1976), 366–389.